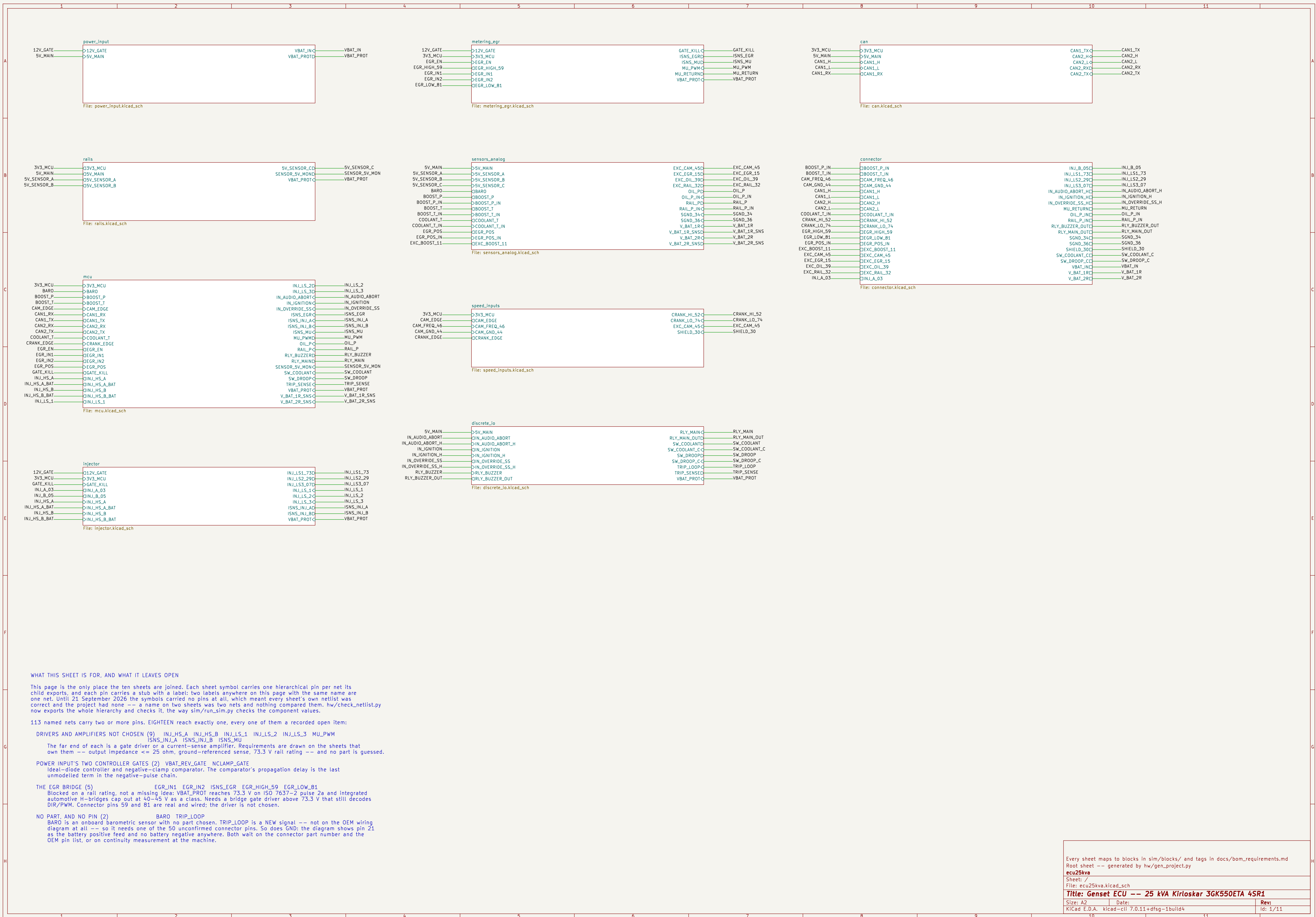
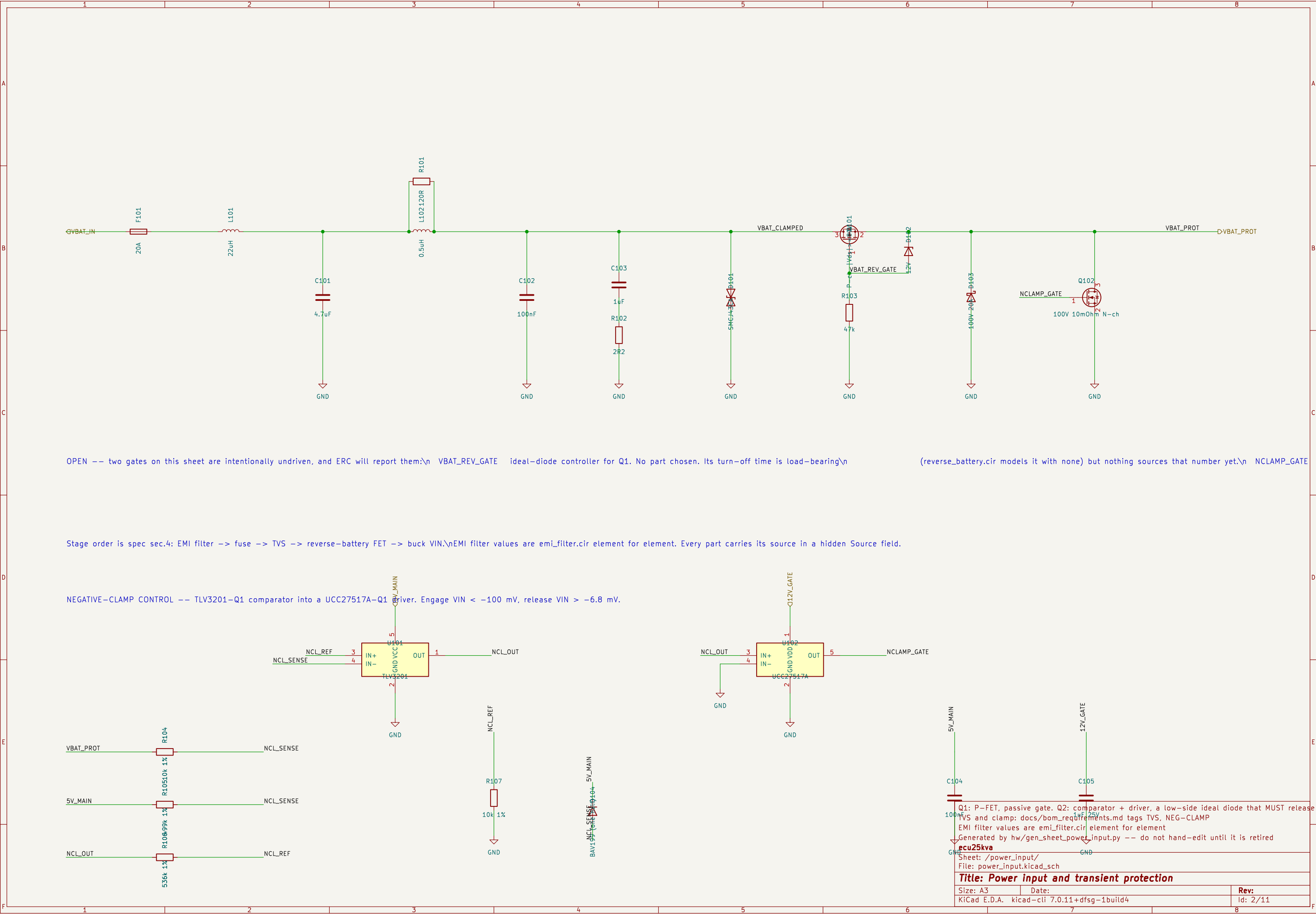






OPEN -- two gates on this sheet are intentionally undriven, and ERC will report them:\n VBAT_REV_GATE ideal-diode controller for Q1. No part chosen. Its turn-off time is load-bearing\n (reverse_battery.cir models it with none) but nothing sources that number yet.\n NCLAMP_GATE comp

Stage order is spec sec.4: EMI filter -> fuse -> TVS -> reverse-battery FET -> buck VIN.\nEMI filter values are emi_filter.cir element for element. Every part carries its source in a hidden Source field.

NEGATIVE-CLAMP CONTROL -- TLV3201-Q1 comparator into a UCC27517A-Q1 driver. Engage VIN < -100 mV, release VIN > -6.8 mV.

Q1: P-FET, passive gate. Q2: comparator + driver, a low-side ideal diode that MUST release

TVS and clamp: docs/bom_requirements.md tags TVS, NEG-CLAMP

EMI filter values are emi_filter.cir element for element

Generated by hw/gen_sheet_power_input.py -- do not hand-edit until it is retired

ecuc25kva

Sheet: /power_input/

File: power_input.kicad_sch

Title: Power input and transient protection

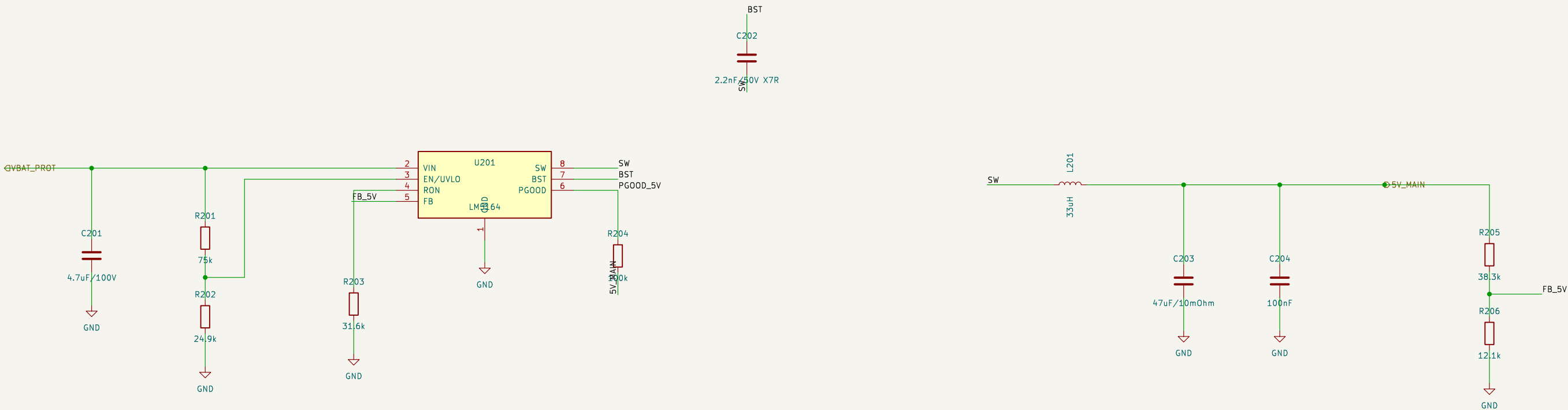
Size: A3

Date:

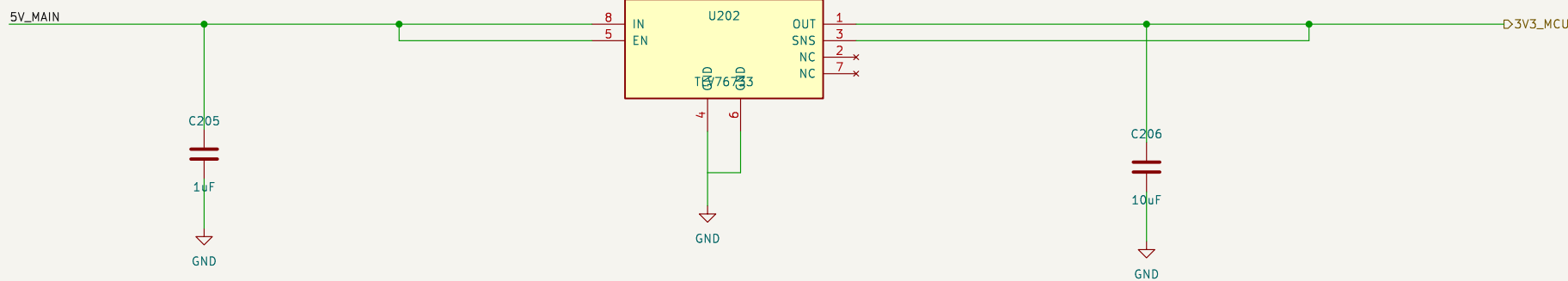
Rev:

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

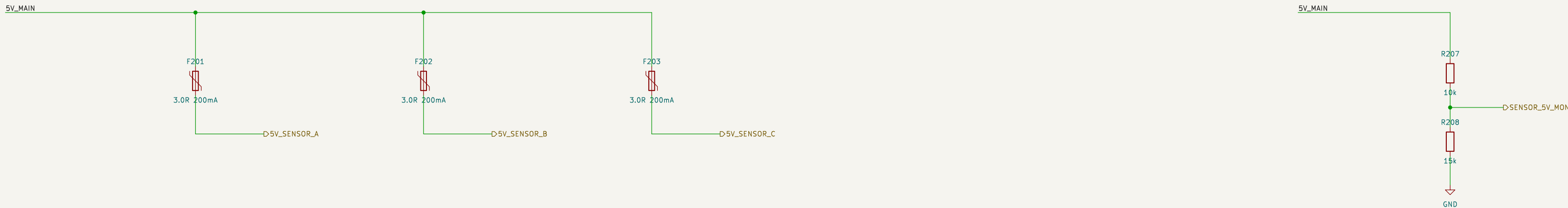
Id: 2/11



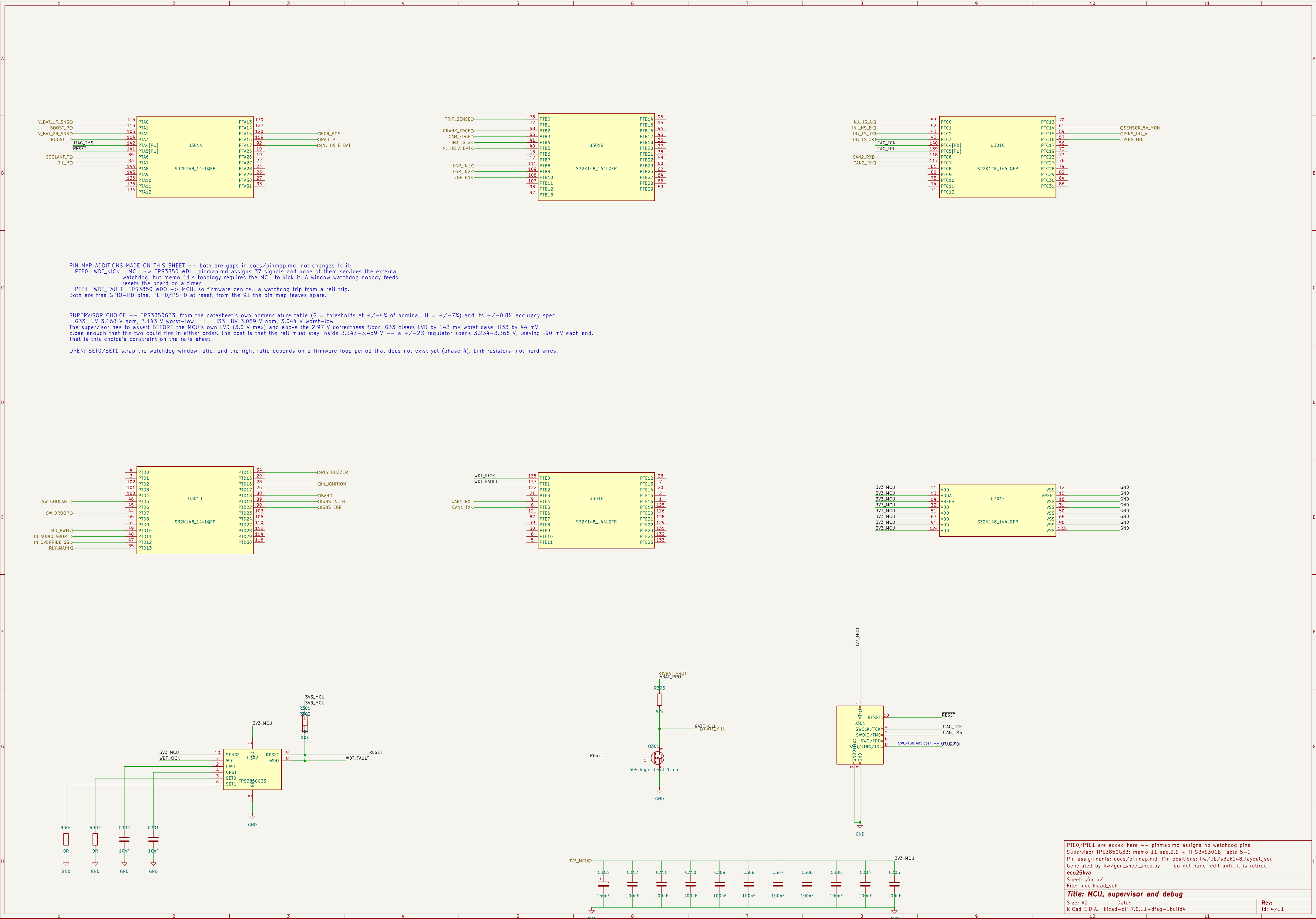
TWO PARTS CHOSEN HERE, both replacing research memo 07 entries that were marked representative rather than chosen:
LM5164 memo 07 sec.4 names a 60 V TPS54360B-Q1. transient_clamp.cir now carries a PASSING check that reads "60 V-class parts are NO LONGER viable downstream" -- raising the TVS standoff to 43 V, so the part stops conducting during a NORMAL clamped dump, put the pulse 2a clamp at 73.3 V. Every rating check in the suite has been written against the LM5164's own -0.3 V / 100 V VIN limits all along; the BOM was simply never reconciled with it. A 60 V buck here is a part that fails the first pulse 2a event.
TLV76733 memo 07 sec.6 names TLV1117-33QDCYRQ1, explicitly "class pricing, not fetched". The supervisor chosen on the mcu sheet made 3V3 accuracy load-bearing: outside 3.143-3.459 V the board resets itself. TLV767-Q1 is 1% over load AND temperature, giving 3.267-3.333 V.



RATIOMETRIC MONITOR, a limitation worth stating: the divider taps 5V_MAIN, upstream of the PTCs. Each group's own PTC drops about 50 mA x 3.0 ohm = 150 mV, which is 3% of the rail and is NOT captured by a measurement taken before it. Correct for the regulator's drift, not for the protection element's. Moving the tap downstream would fix one group and leave the other two wrong, so the tap stays common and the limit is recorded here.

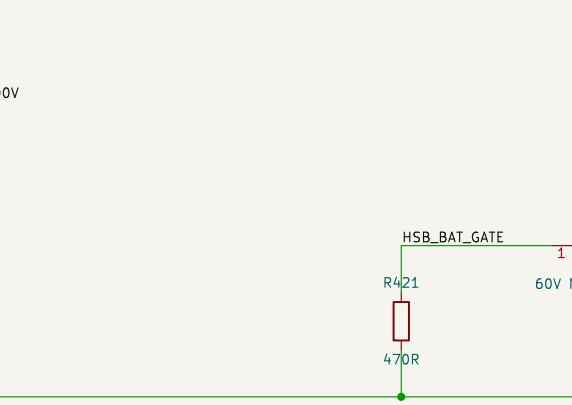
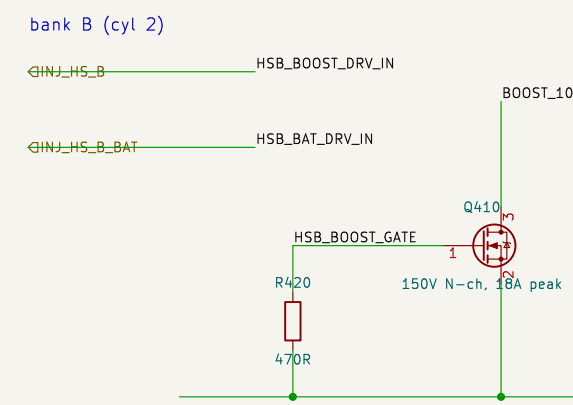
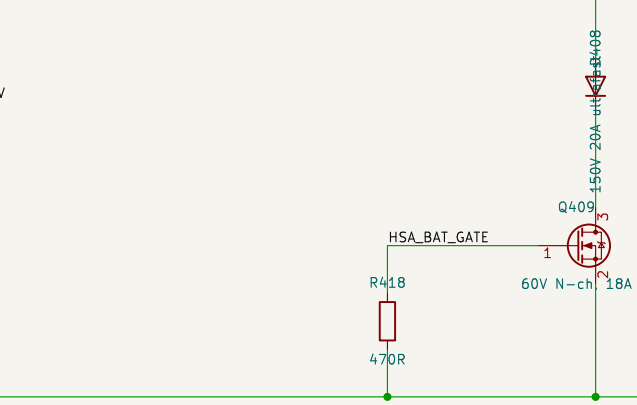
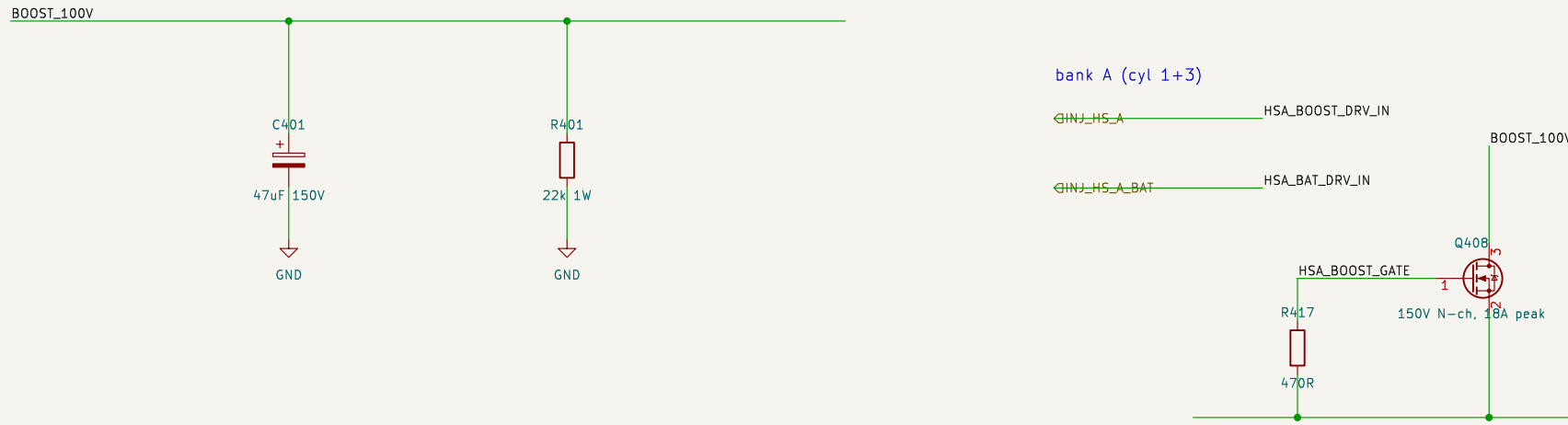


PGOOD_5V is brought out and left unassigned. The LM5164 asserts it open-drain; the pin map has no channel for it. Available, not wired to anything -- shown rather than hidden.

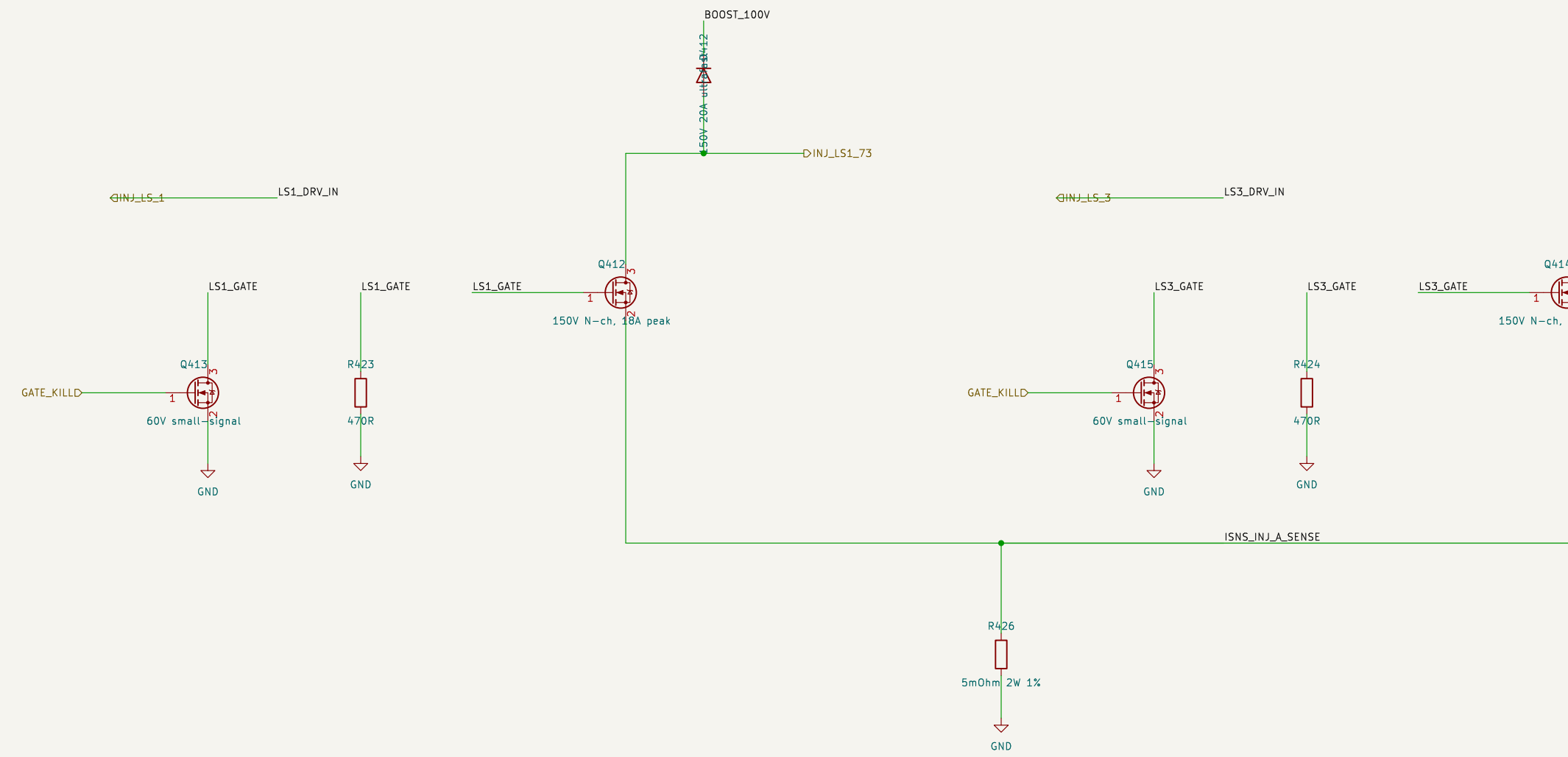


BOOST RAIL -- 100 V reservoir for the injector peak phase

the converter that feeds this rail is drawn at the foot of the sheet -- TP540210-Q1. VBAT_PROT to BOOST_100V



GATE_KILL reaches the four high sides at the drivers' HIN inputs, not at the gates -- a ground-referenced kill FET cannot short a gate floating 100 V up, but HIN is ground-referenced. See the gate-driver block on the right.



A GROUND-REFERENCED KILL CLAMP CANNOT SERVICE A HIGH-SIDE GATE, and spec sec.4's fail-safe list names pins 02/05 alongside the four low-side gates as though it could. The three low sides here get exactly the network supervisor.cir sized and docx/pinmap.md sec.2 confirmed pin by pin: a 470 ohm pulldown to ground, plus a small-signal FET that shorts the gate to ground when GATE_KILL asserts, with no MCU involvement -- which is what makes it work against a hung MCU and not only a reset one.

The FOUR high sides cannot use it -- two per bank since the 21 Sep correction. The source of a high-side N-FET swings to the boost rail, so a pulldown to GROUND would hold Vgs at minus the source voltage and fight the driver on every edge, and a kill FET referenced to ground cannot short a gate that's floating 100 V up, what is drawn instead is the 470 ohm across GATE and SOURCE -- the same Miller-coupling divider supervisor.cir sized, just referenced to the terminal it has to be referenced to -- and the kill path moves into the driver; GATE_KILL must reach the high-side driver's LOGIC INPUT. AUIRS2181S has no shutdown pin, so the same kill FET sits on HIN behind a 1k from the MCU -- the one ground-referenced point in that path. That is drawn in the gate-driver block, and it is the sixth finding of the shape this project keeps turning up: a document and an executable file agreeing with each other and both wrong about a case neither modelled.

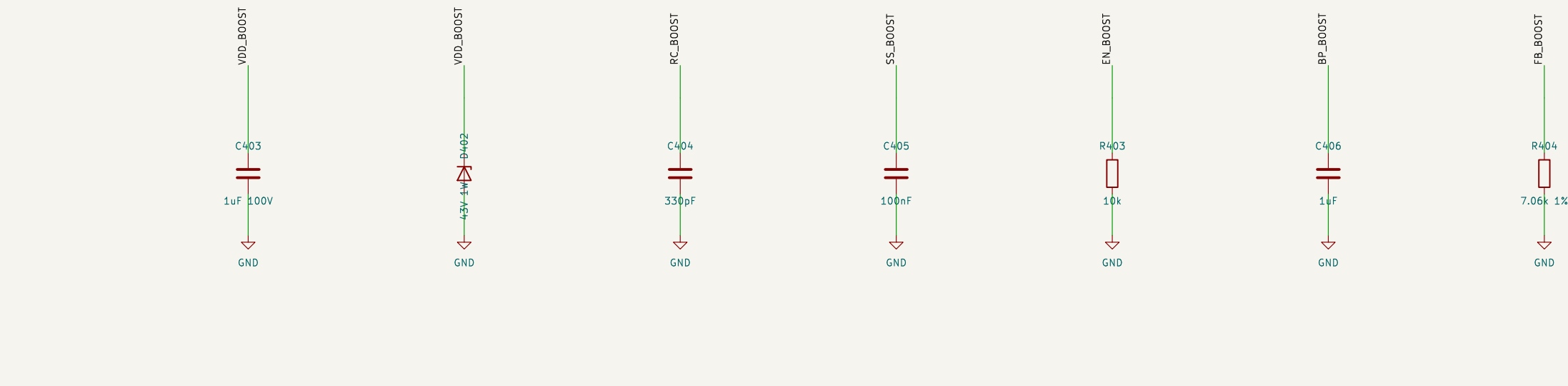
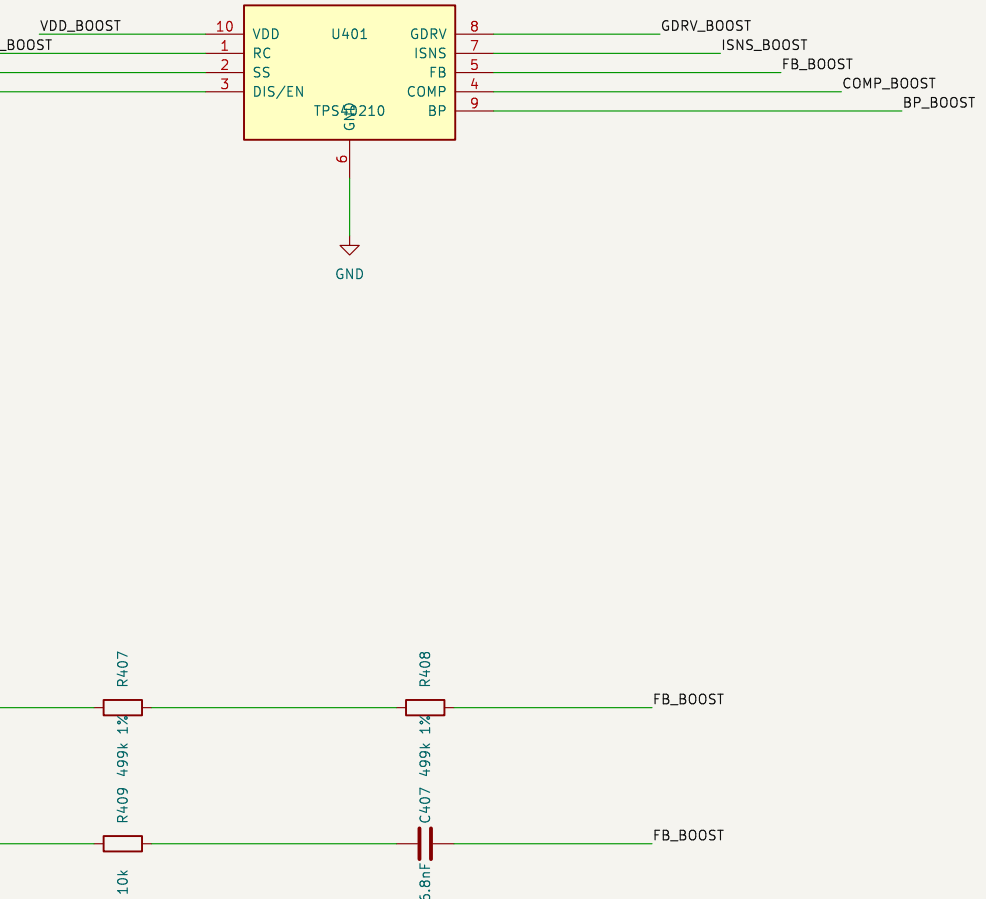
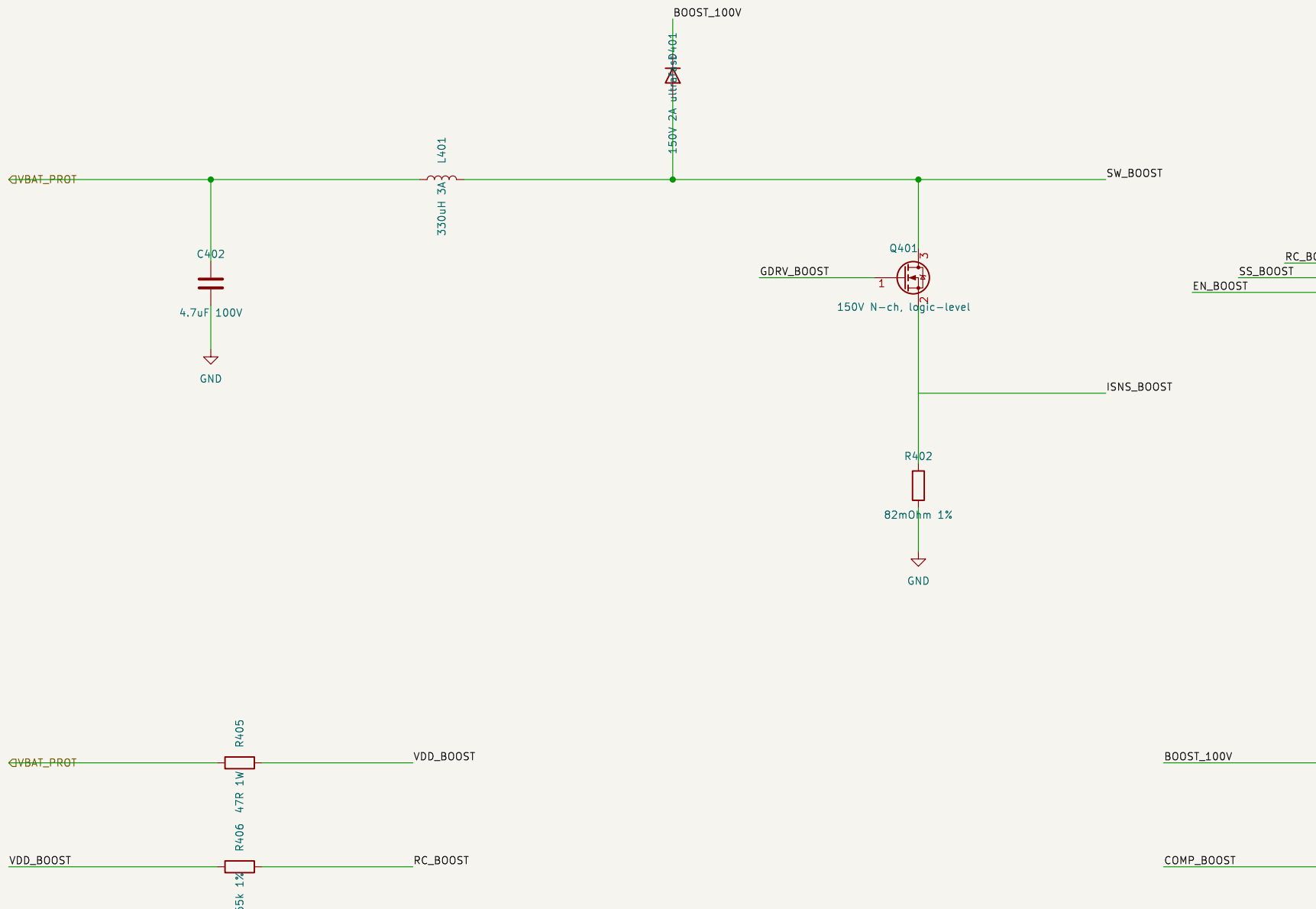
GATE DRIVERS: AUIRS2181S x4, chosen 21 Sep 2026. Its first listed application is 'Piezo / common rail injection', and three properties decided it, each against a part that failed on it. NO CROSS-CONDUCTION INTERLOCK: in a half bridge HO and LO together is shoot-through, but here the high and low switches are in SERIES with the coil and must BOTH be on to fire it -- UCC27282-Q1 and UCC27212-Q1 have interlock and could not fire an injector at all. 600 V OFFSET, against UCC27211A-Q1's 120 V HB absolute maximum, which the bank node plus 12 V clears by 4 V. VS OPERATIONAL TO -5 V, against UCC27211A-Q1's -1 V HS DC minimum, which D_in violates on every hold off-time at 1.0-1.5 V. Each package plus one high side with one low side; the pairing is by package, not by circuit, and UB2's low side is spare.

12V_GATE: the board had no rail between 5 V and 40 V, and the drivers need 10-20 V, it is made from the BOOST rail, because a regulator from the battery cannot make 12 V from a 6-9 V cranking battery -- and a gate rail that collapses during cranking means no injection exactly when the engine needs it. Not the LMS16A already on the board: its VIN absolute maximum is 100 V and this rail sits at -102 V. A discrete follower with a current limit, for a -3 mA load. If the boost stops, every driver drops into UKO and every gate turns off -- the failure runs the safe way.

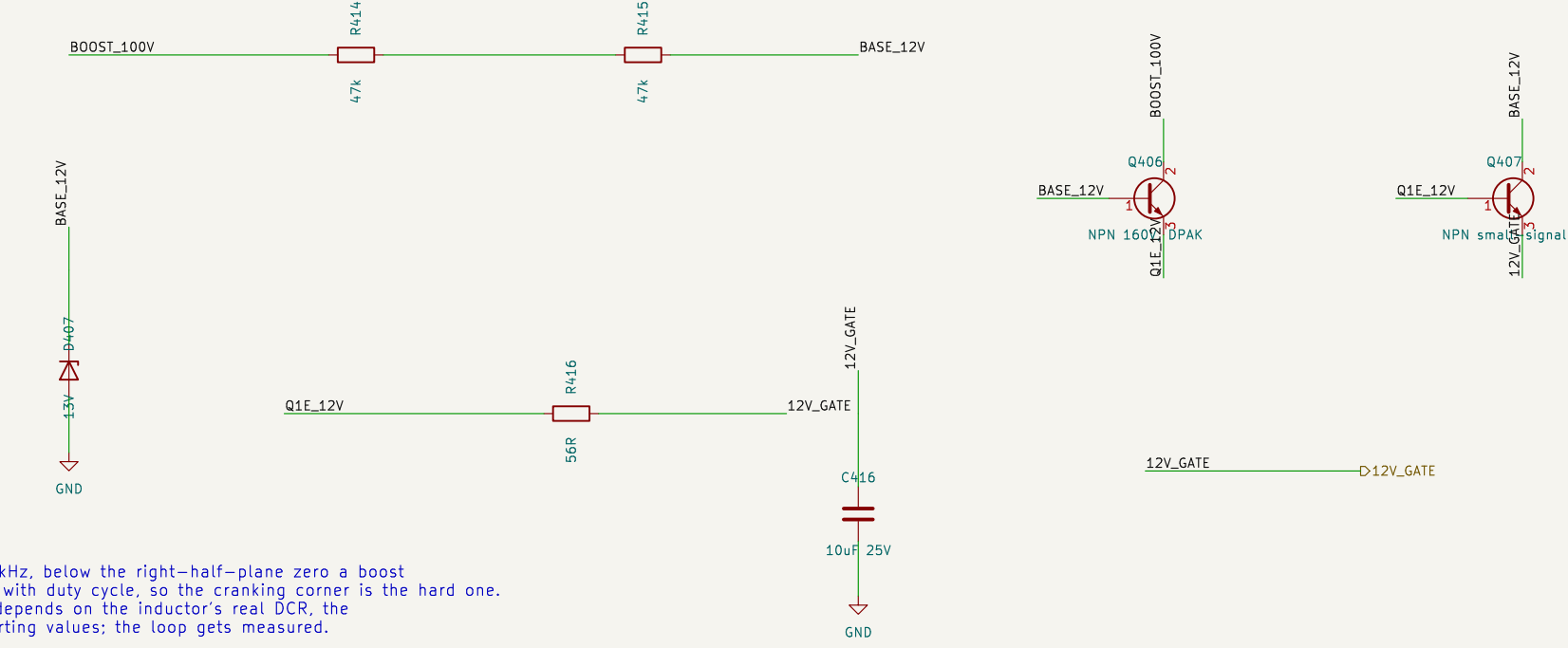
STILL NOT CHOSEN, the two current-sense amplifiers (ISNS_INJ_A/B_SENSE --> PTC15 / PTD19). Ground-referenced -- that is why the shunts went low-side -- gain about 33, 100 kHz or better, and a Kelvin tap at the shunt pad.

THE BOOST CONVERTER IS NOW DRAWN, AND THE PART WAS CHOSEN ON ONE NUMBER; IDFF(min) = 200 ns max, TI SLY5861P Table 6.6. A boost needs 94.1% duty at the 6 V cranking corner, and at 150 kHz that off-time floor allows 97.0%. Most wide-input boost controllers in this class switch at 2.2 MHz, where the same 200 ns caps duty at 56% -- below even the 86.6% the rail needs at a nominal 13.5 V battery. Switching SLOWER is the requirement here, which runs the opposite way to the usual 'higher frequency, smaller magnetics'.

THE RATING THAT DOES NOT FIT, and it is the third time this board has hit it. VDD absolute maximum is 52 V (Table 6.1) and VBAT_PROT reaches 73.3 V for about 50 us on ISO 7637-2 pulse 2a -- over by 1.4x. Same shape as DRV8873-Q1's 40 V VM, which took that part off the EGR bridge. It is solvable here for a reason that did not apply there: VDD draws 2.5 mA plus gate charge, not a motor's current, so 47 ohm and a 43 V zener hold the pin inside its rating through the pulse, 43 V rather than 39 V so the clamp stays OUT of conduction during the 40 V load dump, which lasts 400 ms instead of 50 us and would otherwise sit it at 0.8 W. The POWER stage is untouched: the inductor and FET see the rail directly and are 150 V class for it. All of this is checked in run_sim.py under boost_converter -- not with a switching model, which that block deliberately does not have, but as executable arithmetic on the datasheet's own ratings, the same treatment transient_clamp.cir gives the 73.3 V rail. BOOST CONVERTER -- TP540210-Q1, 6-40 V in, 100 V out at 50 mA average. The reservoir above delivers the pulse; this delivers the trickle.



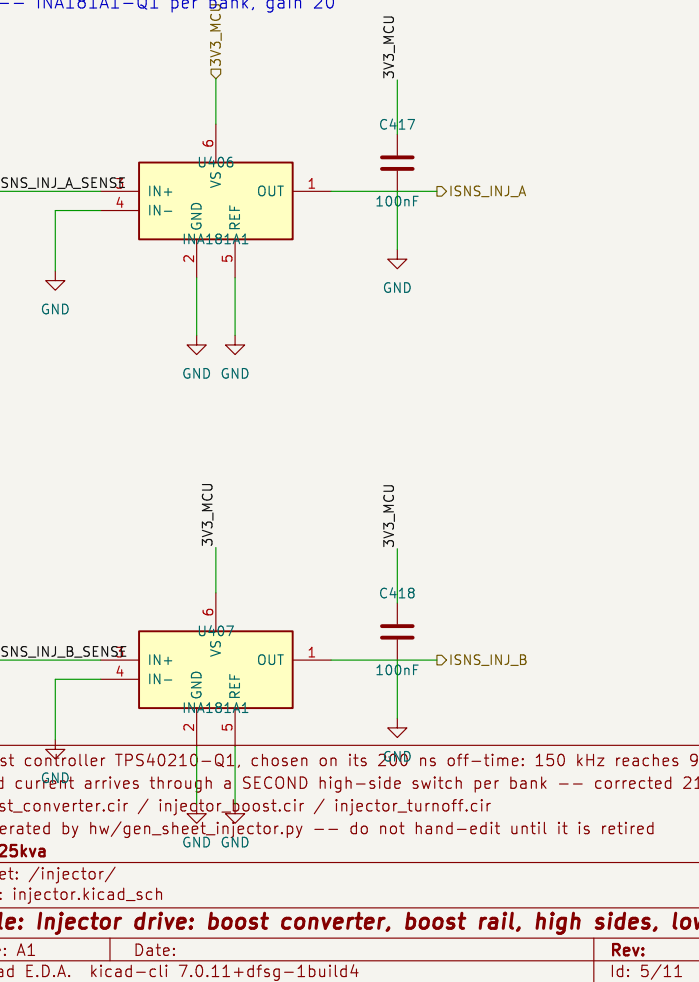
12V_GATE -- gate-drive supply for every driver on the board, from the boost rail so it survives cranking



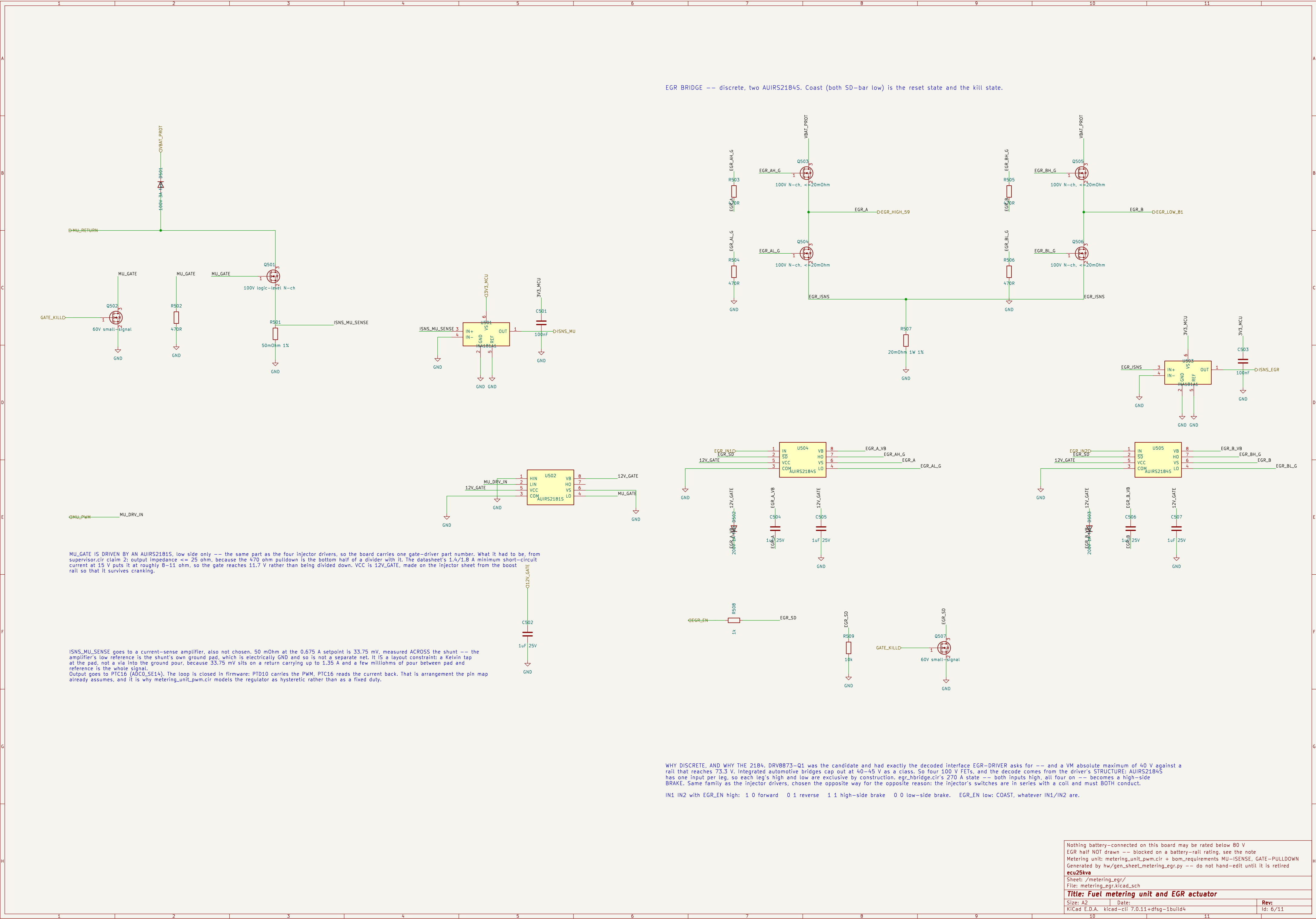
THE COMPENSATION VALUES ARE A STARTING POINT AND THE SHEET SAYS SO. Rcomp 10k and Ccomp 6.8 nF put the error amplifier's zero at 2.3 kHz, below the right-half-plane zero a boost converter puts at (1-D)/2 * Rload / (2*pi*L) = 17.3 kHz at 13.5 V in -- which is the constraint that actually bounds crossover, and it MOVES with duty cycle, so the cranking corner is the hard one. Everything else on this sheet traces to a datasheet number or a simulation result. These two do not, and cannot; a current-mode boost's loop depends on the inductor's real DCR, the capacitor's real ESR and the FET's real switching behaviour. They are bench measurements, what is defensible now is the TOPOLOGY and the starting values; the loop gets measured.

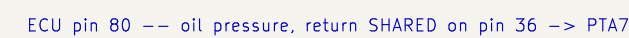
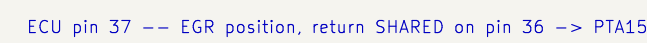
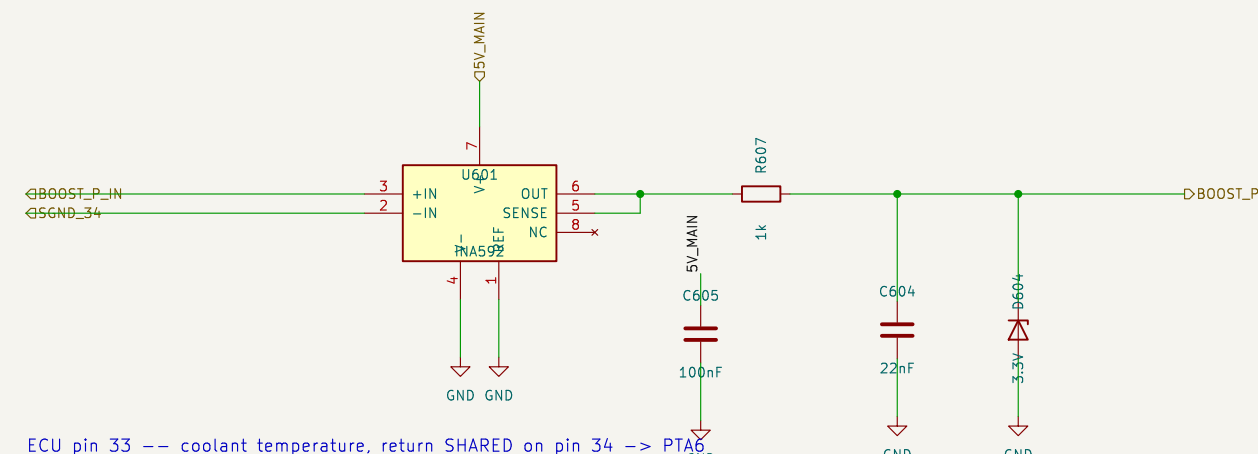
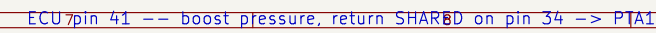
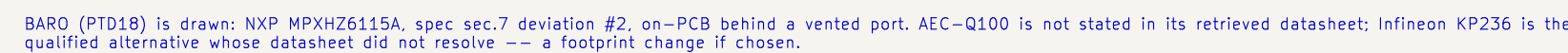
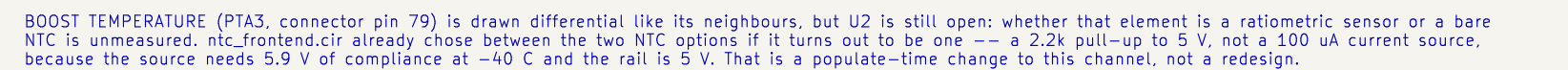
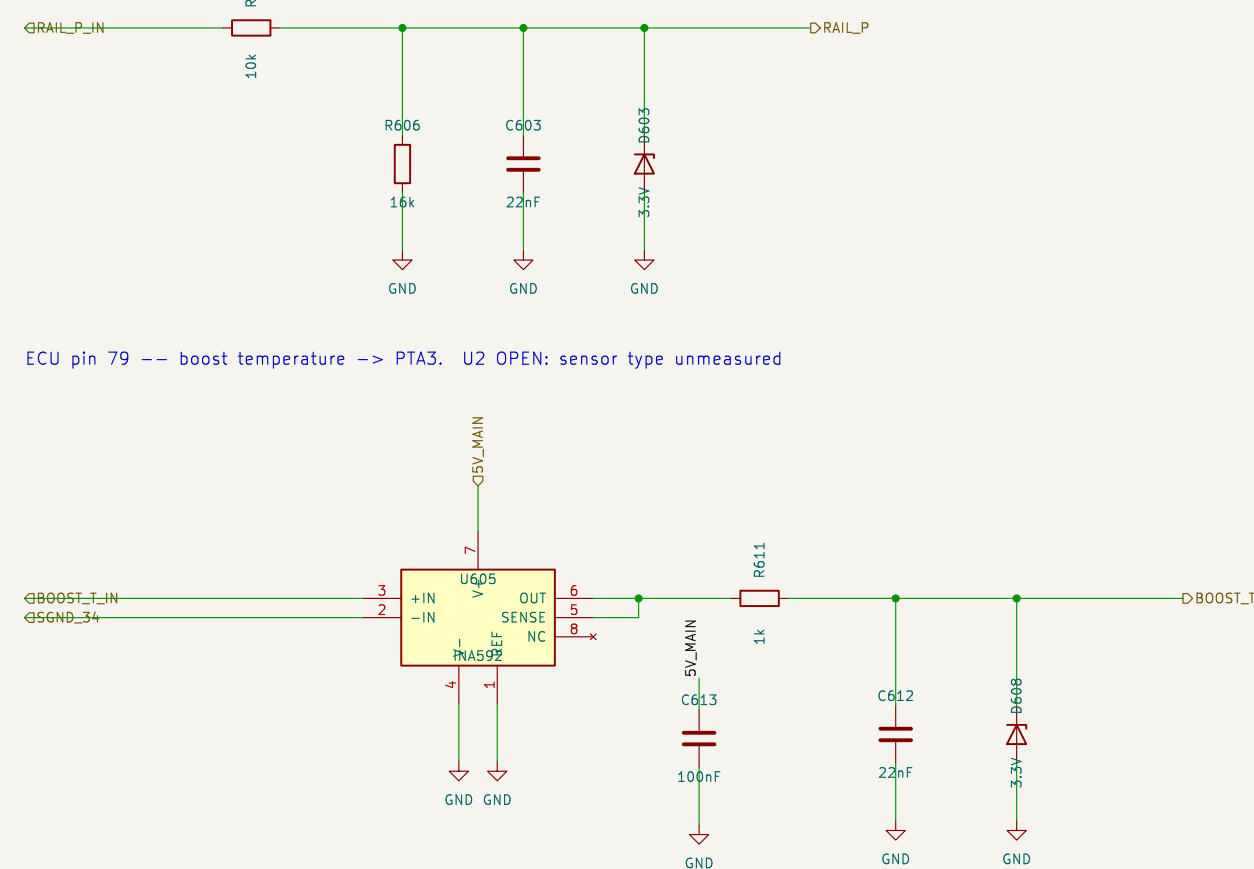
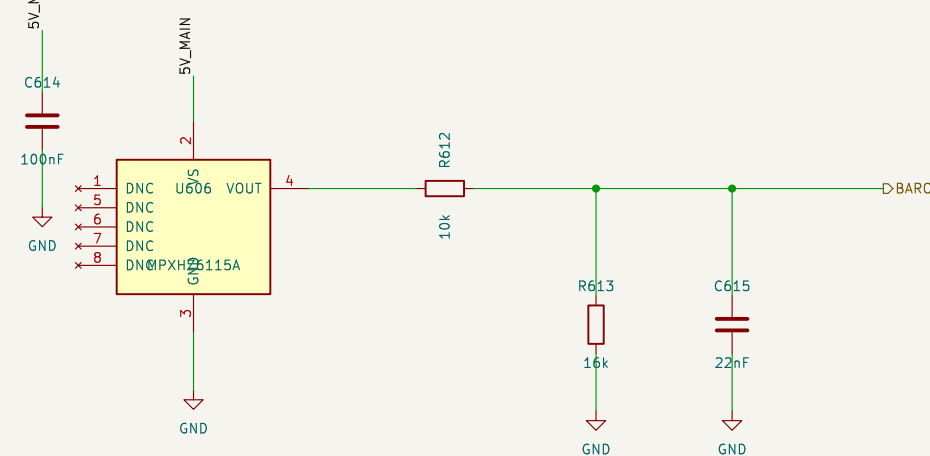
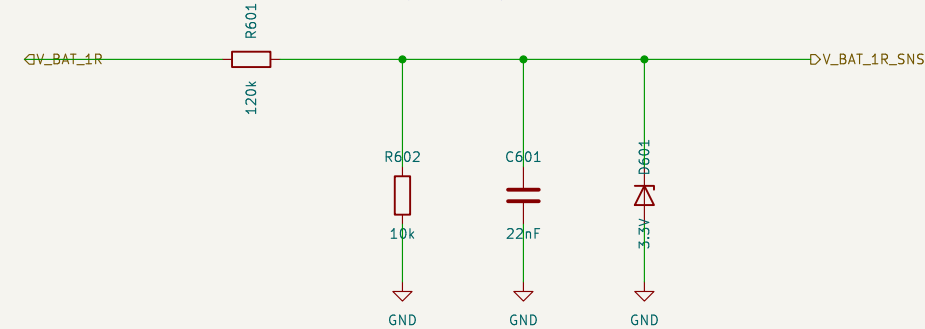
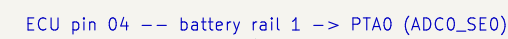
NOT DRAWN, and deliberately: no output capacitor appears here, because the 47 uF reservoir at the top of this sheet IS the output capacitor. Its 22k bleed is the only thing that can take charge OFF this rail -- a boost converter cannot pull its own output down, which is the same one-sided regulation boost_converter.cir's arrangement C measures.

CURRENT SENSE -- INA181A1-Q1 per bank, gain 20

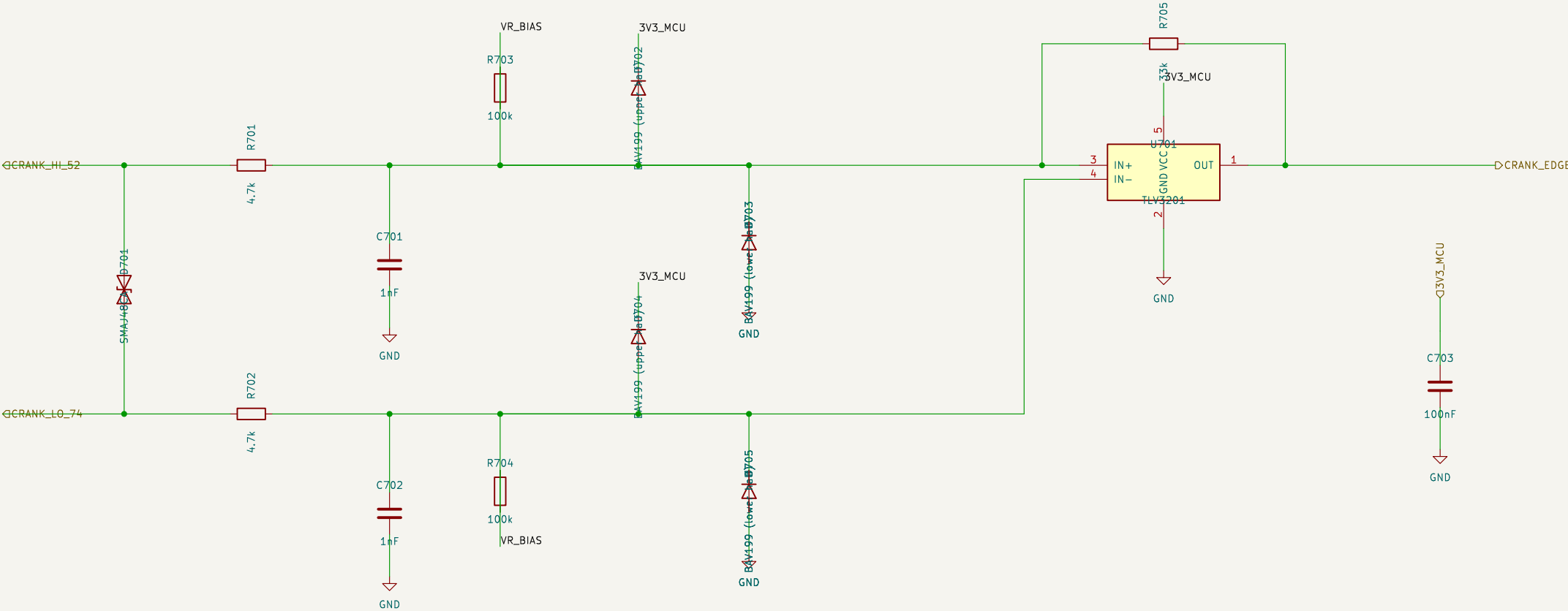


Boost controller TP540210-Q1, chosen on its 200 ns off-time; 150 kHz reaches 94% duty at cranking. Hold current arrives through a SECOND high-side switch per bank -- corrected 21 Sep 2026. boost_converter.cir / injector_boost.cir / injector_turnoff.cir. Generated by hw/gen_sheet/injector.py -- do not hand-edit until it is retired. eu29kva. Sheet: /injector/. File: injector.kicad_sch. Title: injector drive: boost converter, boost rail, high sides, low sides. Size: A1. Date: Rev: 16/ 5/11. Kicad E.D.A. kicad-cir 7.0.11+dfsg-1build4

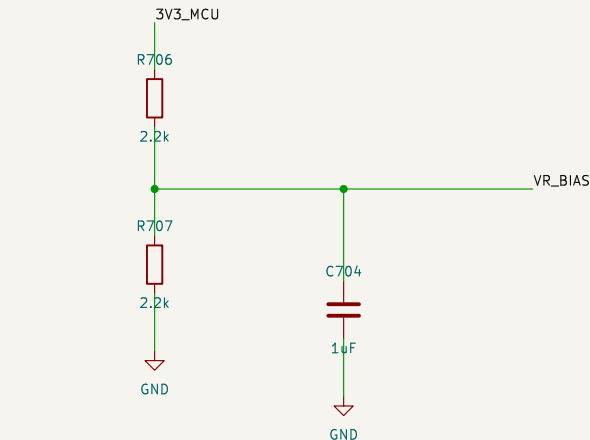




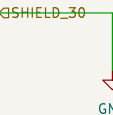
CRANK -- variable reluctance, ECU pins 52 / 74, shield 30 -> zero-cross comparator -> PTB2 (FTM1_CH0)



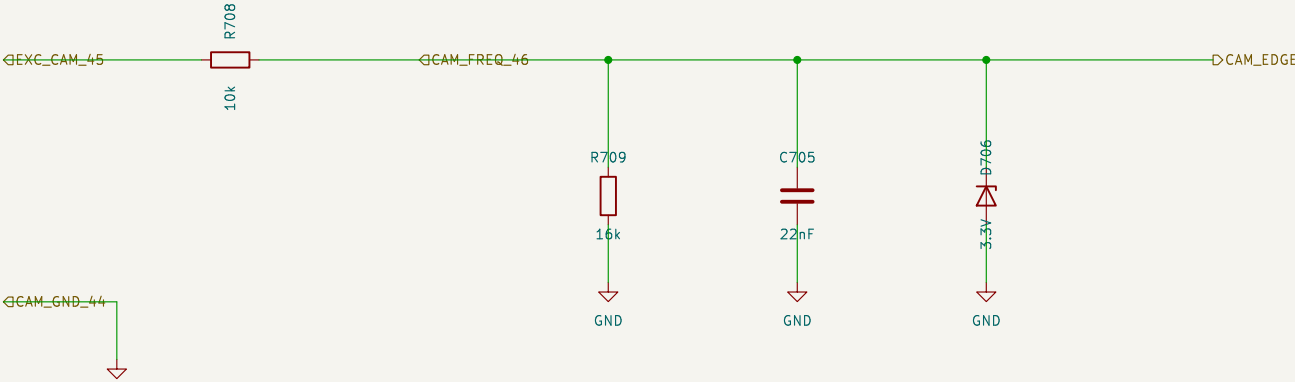
VR_BIAS -- mid-rail reference for the floating VR coil



ECU pin 30 -- crank sensor shield, terminated at THIS end only. Grounding both ends makes the shield a second conductor between two grounds that differ by the injector return current, and that current then flows down the shield around the pair it is there to protect.



CAM -- digital Hall, ECU pins 45 / 46 / 44 -> PTB3 (FTM1_CH1). CORRECTS the front-end spec sec.4 specifies literally



CLAMP LEAKAGE IS THE BINDING SPEC, WHICH INVERTS THE RULE THIS PROJECT USED ON THE POWER PATH. Every earlier clamp on this board was chosen for forward drop -- the negative-clamp Schottky's 0.547 V at 15 A decomposes into junction and bulk terms and that decomposition is what ruled out paralleling junction devices. Here the forward drop does not matter at all: these diodes conduct only on a fault, into a 4.7k series resistor that is already doing the limiting. What matters is what they do when they are OFF.

A reverse-biased diode's leakage flows out of the 4.489k Thevenin at the input node and appears as an offset voltage there. The hysteresis window is +/-198 mV, so:

BAT54S--class Schottky -2 uA at 25 C, and Schottky leakage rises roughly 100x to 125 C -> -200 uA -> 0.90 V of offset.
That is 4.5x the whole hysteresis window. The comparator latches and the engine has no speed signal at temperature.
BAV199--class low-leakage silicon 5 nA max at 25 C, under 1 uA at 125 C -> under 4.5 mV of offset, 2% of the window.

The two clamp diodes are biased symmetrically (both reverse-biased by 1.65 V from VR_BIAS) so their leakages partly cancel, and both legs are matched so what survives is partly common-mode -- but a design whose speed input depends on cancellation between two diodes' hot leakage is not one to defend. The requirement is stated instead: reverse leakage at or below 1 uA at 125 C. vr_conditioner.cir does NOT model this -- its comparator is an ideal switch with no input network at all, so no clamp, no leakage and no offset exist in it. This is the sheet finding something the block could not.

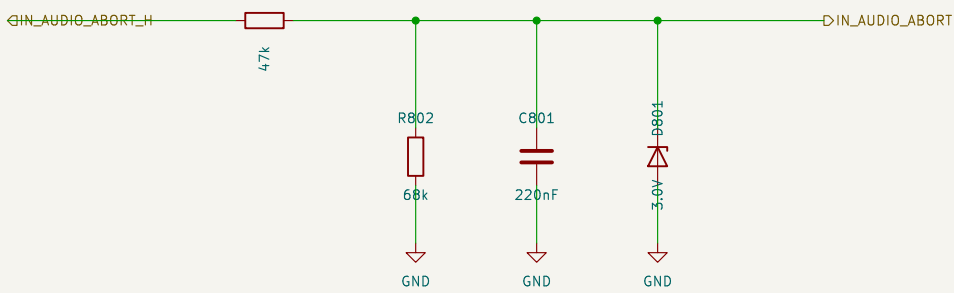
WHY THE CAM FRONT-END ON THIS SHEET IS NOT THE ONE THE SPEC ASKS FOR. Spec sec.4 and docs/pinmap.md sec.1.3 both give pin 46's front end as "Pull-up to 5V_SENSOR -> timer capture": the Hall output pulled to 5 V and taken straight into PTB3. NXP S32K1xx Data Sheet Rev. 15, Table 17, puts Vih max at VDD + 0.3 V, about 3.6 V at VDD = 3.3 V, so a full 5 V swing is not a guaranteed valid logic level. Table 1 does allow -0.8 to +5.8 V on any I/O pin, but only while respecting the +/-3 mA injection limit, and the datasheet's own words are that "functional operation at the maximum values is not guaranteed" -- a stress margin, not a design basis for a pin that would sit at 5 V once per cam revolution for the machine's running life. The 16k to ground turns it into the same 10k/16k divider the ratiometric sensor channels already use. It costs one resistor.

This was found by writing cam_frontend.cir, which pinmap sec.4 asked for by name before this sheet was drawn: no block had ever simulated a cam front-end -- vr_conditioner.cir covers the crank and there was no cam equivalent -- so a 5 V swing on a 3.3 V pin had never been checked. Fourth finding of the same shape in this project: a document and an executable file disagreeing, with nothing comparing them.

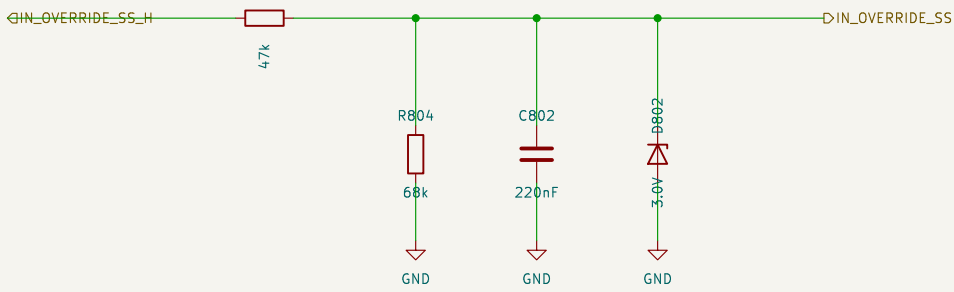
PARTS: the comparator (now TLV3201--Q1, the clamp comparator on power_input) and the cam Hall sensor itself, which is U8 on the unknowns register and needs the machine. The comparator's requirement is unusually easy, and deliberately so: biasing the floating coil to VR_BIAS means the inputs never leave mid-rail, so no part is needed whose common-mode range includes ground -- the usual hard constraint on a zero-cross detector. Single 3.3 V supply, push-pull rail-to-rail output, propagation delay at or below 1 us (0.009 crank degrees at rated speed on a 60-tooth wheel). Input offset voltage is a non-issue against a 198 mV window.

Clamp diodes specified by REVERSE LEAKAGE, not forward drop -- see the sheet note		
Zero-cross with +/-198 mV hysteresis: a fixed threshold cannot see a 2 V cranking signal		
vr_conditioner.cir / cam_frontend.cir		
Generated by hw/gen_sheet_speed_inputs.py -- do not hand-edit until it is retired		
ecu25kva		
Sheet: /speed_inputs/		
File: speed_inputs.kicad_sch		
Title: Crank VR conditioner and cam front-end		
Size: A2	Date:	Rev:
KiCad E.D.A.	kiCad-cli 7.0.11+dfsg-1build4	Id: 8/11

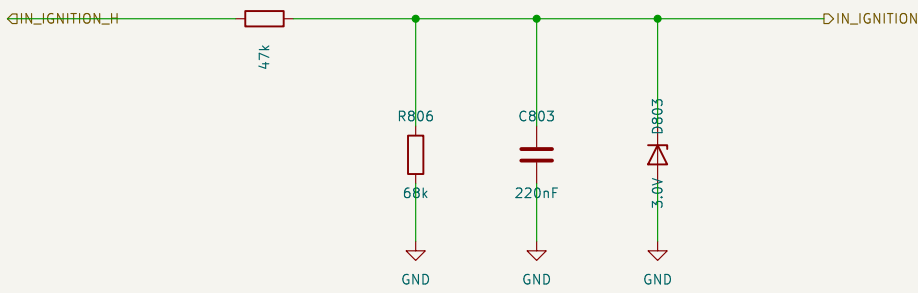
ECU pin 20 -- audio abort, active high -> PTD11



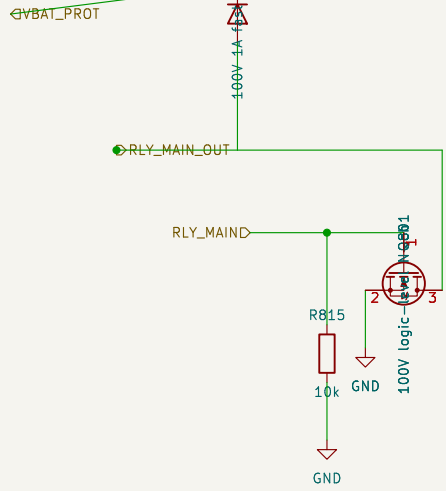
ECU pin 24 -- override start/stop, active high -> PTD12



ECU pin 71 -- ignition, active high -> PTD16



ECU pin 50 -- main relay, low side <- PTD13

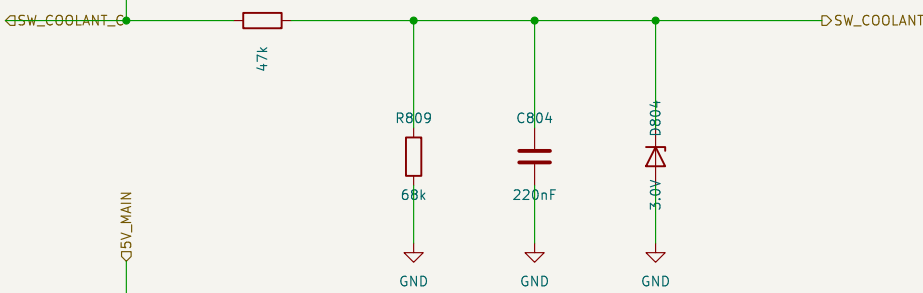


TRIP SENSE MOVES FROM PTD6 TO PTB0 (ADC0_SE4). docs/pinmap.md assigns it as a GPIO input; trip_module_sense.cir designs a THREE-STATE supervised loop and checks the three bands with real margins -- healthy 2.863-2.954 V, tripped 0.310-2.068 V, wire fault 0.000 V. A GPIO reads two states. It separates healthy from not-healthy and nothing else, which discards exactly the distinction the bleed resistor exists to create: a supervised loop that cannot report a cut wire is an unsupervised loop with extra parts. PTB0 is free, bonded out on the 144-LQFP, and Hi-Z with no pull at reset. PTD6 returns to the spare pool.

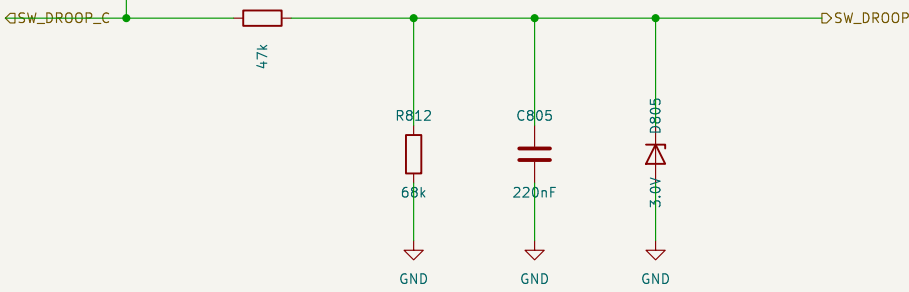
The 1.2 Mohm bleed resistor that makes the tripped band is INSIDE the trip module, across its contact -- it is not on this board and is not drawn here. It is also INFERRED rather than measured (trip_module_sense.cir says so), so the band edges above move if the real part differs.

WHY THE SAME FRONT-END ON EVERY CHANNEL. The 47k/68k/220nF/3.0V chain is discrete_input.cir's, and it is used unchanged on all six inputs -- including the two whose switch is a contact with real rather than a feed. Those two get a 10k pull-up BEFORE the 47k, so a battery short on the harness still meets the clamp chain rather than the pull-up. An open contact then leaves the divider at 2.72 V, under the 3.0 V knee, so it reads high without the zener conducting.

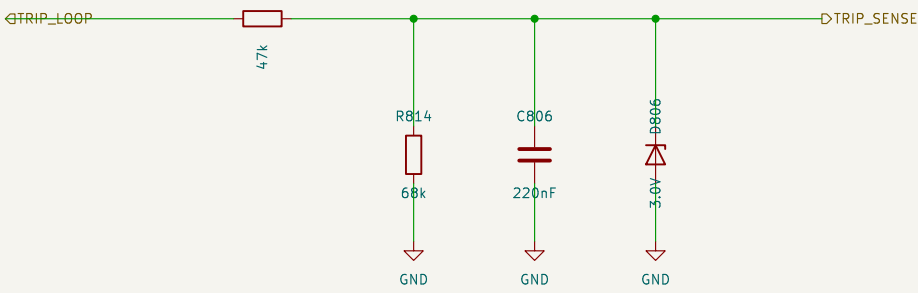
ECU pin 43 -- coolant switch, contact to ground -> PTD5



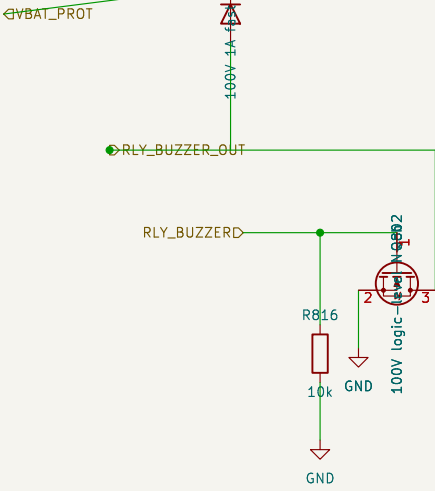
ECU pin 23 -- droop switch, contact to ground -> PTD7



NEW pin -- supervised trip loop -> PTB0 (ADC0_SE4), not PTD6

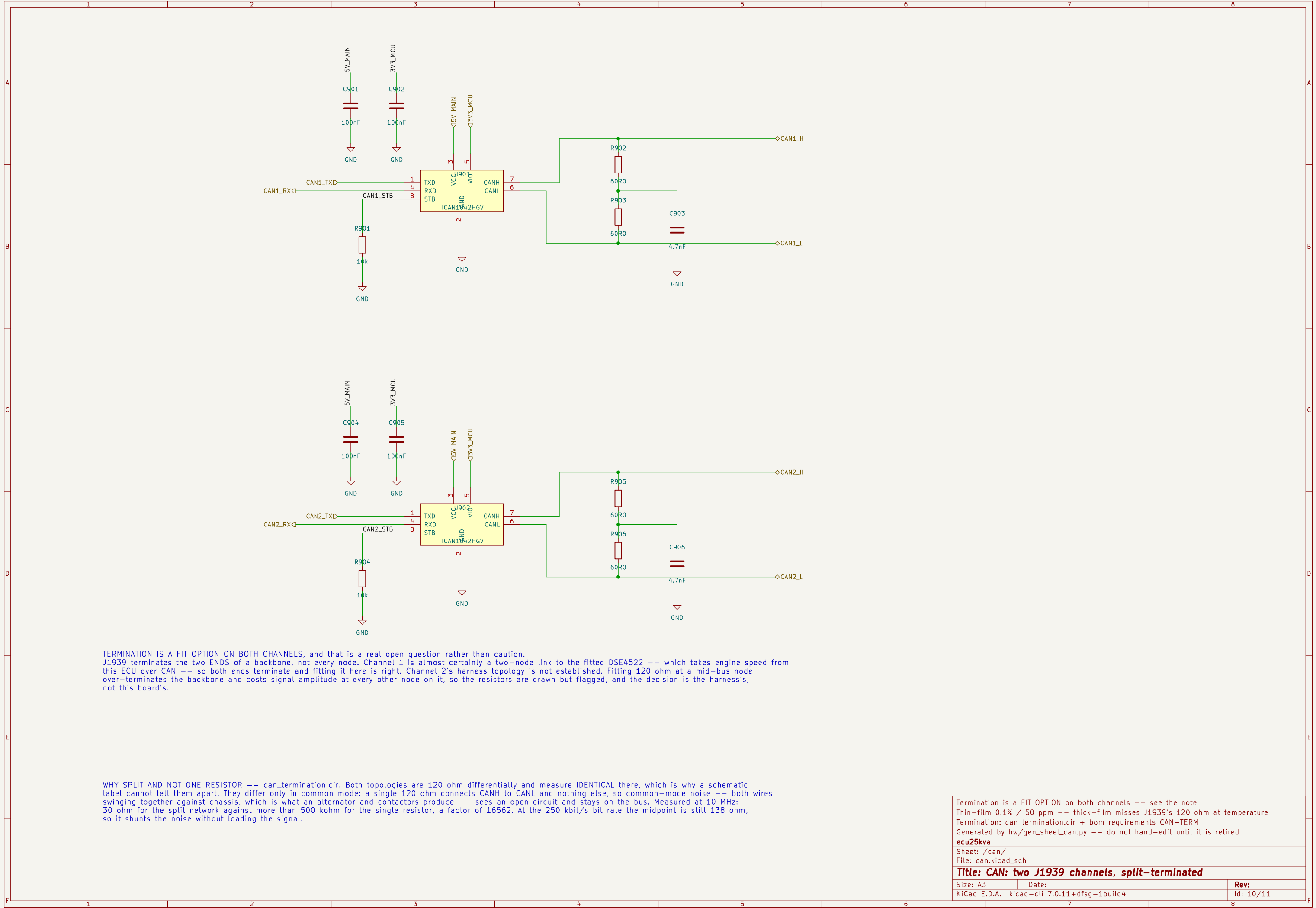


ECU pin 69 -- buzzer relay, low side <- PTD14



Relay FETs are 100 V class -- VBAT_PROT reaches 73.3 V on pulse 2a
Trip sense moved PTD6 -> PTB0: a GPIO cannot read a three-state loop
Front-ends: discrete_input.cir / trip_module_sense.cir. Outputs: relay_driver.cir
Generated by hw/gen_sheet_discrete_io.py -- do not hand-edit until it is retired

ecv25kva		
Sheet: /discrete_io/		
File: discrete_io.kicad_sch		
Title: Discrete I/O: switch inputs, trip loop, relay drivers		
Size: A2	Date:	Rev:
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 9/11



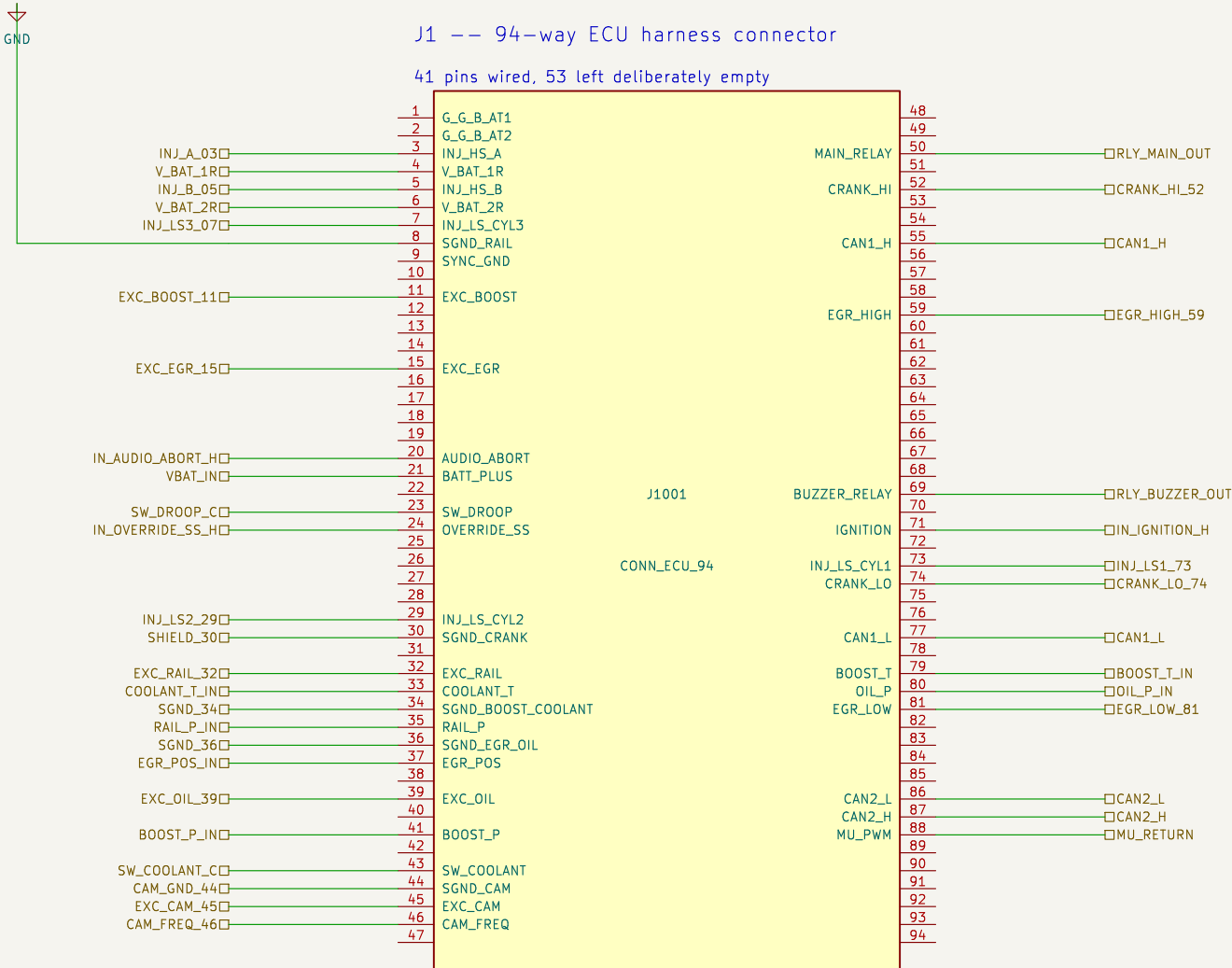
THE POWER GROUND HAS NO PIN, AND THAT IS THIS SHEET'S FINDING.

GP3.8703.C4.pdf shows pin 21 as the battery positive feed and shows no battery negative anywhere on the ECU connector. Its own scope note explains why -- It is a field troubleshooting diagram and it omits "power grounds, unused pins and internal-only pins" -- so this is a gap in the SOURCE, not a contradiction in it.

It is still a gap that stops a board. Every other sheet in this project returns to GND: the injector low sides carry 18 A into it, the metering unit up to 1.35 A, and the analog front ends reference their zeners and their 22 nf filters to it. At this boundary that net has nowhere to go.

It cannot be solved by choosing. Fifty pins have no established function and several of them are certainly grounds -- a 94-way connector feeding an 18 A injector stage does not have one return -- but picking which is exactly the guess this project forbids, and a wrong ground pin is not a rework, it is a harness that has to be remade.

WHAT ACTUALLY SETTLES IT: the connector part number and the OEM pin list, or continuity measurement on the machine between the connector shell and battery negative. The reference names the first as the missing document. The second needs the machine and belongs with U16/U2/U8/U14/U5.



PINS SHOWN ON THE DIAGRAM AND STILL NOT WIRED

09 SYNCHRONIZATION GROUND. Drawn RED on a diagram whose own legend makes red a positive feed, so the name and the colour disagree. U6 on the unknowns register.
01 G_G_B AT1. 02 G_G_B AT2. Memo 03 suspects these are power grounds rather than switch-return legs. If that is right they are part of the answer to the note on the left. No MCU pin is assigned either way, because guessing which would violate the same rule.

NETS THAT EXIST AND HAVE NO PIN YET

TRIP_LOOP the trip-module supervised loop is NEW -- it is not on the OEM diagram at all, so it needs one of the 50 unconfirmed pins. discrete_io draws the front end; the pin waits on the same document the ground does.

EGR_HIGH_59 wired here, because pins 59 and 81 ARE on the diagram -- but the bridge that drives them is the blocked half of metering_egr, waiting on a gate driver rated above the 73.3 V this board's own battery rail reaches on ISO 7637-2 pulse 2a. The pins are real; the driver is not chosen.

WHY PINS 34 AND 36 ARE SIGNALS AND PIN 08 IS GROUND

34 and 36 are SHARED sensor returns -- 34 for boost pressure and coolant temperature, 36 for EGR position and oil pressure. A shared return carries both sensors' current, so its own resistance is an offset on both readings, which is why sensors_analog carries them into difference amplifiers rather than to ground. 08 is rail pressure's DEDICATED return: nothing to reject, so it is simply ground here.

THE PART IS IDENTIFIED; THE LAND PATTERN IS NOT. refs/field-evidence-2026-09-18.md sec.2b closed U1 off a photograph of the housing angled to the light; BOSCH and the armature-in-circle logo moulded on the side body, 1 928 405 194 under the BOSCH text, 1 928 405 192 on the lower body strip, CODE C beside it, >PA66-GF50< on the locking lever. Read off the physical part, which beats a catalogue match because it cannot be a mis-identification. CODE C IS LOAD-BEARING: Bosch supplies these housings mechanically coded so differently-coded plugs cannot mate, so any replacement or adapter must match it or it will not engage -- the kind of detail that is invisible until a part arrives and will not fit. Still missing: the OEM PIN LIST, a different document from the part number, and the board-side footprint. A search for both numbers returned no catalogue hit, so the land pattern has to come through a Bosch distributor quoting them. Nothing is assigned here, because a mating land pattern is not a thing to infer from a photograph. Pin NAMES on the symbol are the wiring diagram's own signal names where it shows the pin and KiCad's blank elsewhere -- the same treatment the op-amps and the comparator get: the symbol asserts what is known and declines to assert what is not.

Bosch 1 928 405 192 / 194 code C -- part identified off the housing: land pattern is not			
50 pins have no established function -- including, so far, the power ground			
Pin count and 44 pin functions from GP3.8703.C4.pdf via refs/ecu-pinout-extracted.md			
Generated by hw/gen_sheet_connector.py -- do not hand-edit until it is retired			
ecu25kva			
Sheet: /connector/			
File: connector.kicad_sch			
Title: 94-way ECU connector and harness mapping			
Size: A2	Date:	Rev:	
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 11/11	